SULIT

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION

SEMESTER 2, 2025/2026

SUBJECT CODE	**:**	**SCSM2223**
SUBJECT NAME	**:**	**CROSS-PLATFORM APPLICATION DEVELOPMENT**
SECTION	**:**	**ALL SECTIONS**
TIME	**:**	**(3 HOURS)**
DATE/DAY	**:**	**7 JULY 2026 / TUESDAY**
VENUES	**:**	

INSTRUCTIONS TO THE STUDENTS:

- This exam consists of **THREE (3)** sections with a total of 100 marks: Section A — Multiple-Choice Questions (20 marks), Section B — Structured Questions (45 marks), and Section C — Code Application Questions (35 marks).
- **Answer ALL questions** in the spaces provided in this question booklet.
- You have **THREE (3) HOURS** to complete this exam.
- All code answers must be written using the syntax taught in this course and must be syntactically correct. Comment statements are **NOT** required.
- This is a **CLOSED BOOK** exam. No electronic devices are permitted.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question booklet consists of 15 printed pages including this page)

Mock examination paper generated for personal revision practice — not an official UTM examination paper.

SECTION A: MULTIPLE-CHOICE QUESTIONS [20 MARKS]

(INSTRUCTION: This section consists of TWENTY (20) questions. Each question carries ONE (1) mark. Circle the letter of the BEST answer in this booklet.)

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1. Cross-platform development is best described as:

   A. developing a separate native application for every operating system
   B. developing an application once and running it on multiple operating systems or devices with minimal modification
   C. compiling source code into machine code for one specific device
   D. developing applications that run only inside a single web browser

2. In Java's "Write Once, Run Anywhere" approach, the same bytecode can run on different operating systems because:

   A. the Java compiler produces different machine code for each operating system
   B. Java programs are interpreted directly by the web browser
   C. the bytecode is recompiled by the web server before execution
   D. the Java Virtual Machine (JVM) is available on multiple operating systems

3. Which of the following is a characteristic of a Single-Page Application (SPA)?

   A. The server generates a complete new HTML page for every user action
   B. Every user action causes the browser to reload the entire page
   C. The page loads once and JavaScript dynamically updates parts of the page using data from an API
   D. It is the traditional model used by CGI and JSP applications

4. Which *browser – rendering engine* pairing is CORRECT?

   A. Google Chrome — Blink
   B. Mozilla Firefox — WebKit
   C. Safari — Gecko
   D. Microsoft Edge — Trident

5. Which statement about HTML text semantics is TRUE?

   A. `<b>` conveys importance to screen readers
   B. `<strong>` and `<b>` have exactly the same meaning
   C. `<strong>` conveys importance to screen readers, while `<b>` only draws visual attention
   D. `<em>` is used to make text bold

6. A stylesheet is written *mobile-first*. Which media query correctly ADDS styles for screens 600 px and wider?

   A. @media (max-width: 600px) { ... }
   B. @media (min-width: 600px) { ... }
   C. @media screen and (width: 600px) { ... }
   D. @media only tablet { ... }

7. Which statement selects ALL elements that have the class `card`?

   A. `document.getElementById("card")`
   B. `document.querySelector("#card")`
   C. `document.getElementByClassName("card")`
   D. `document.querySelectorAll(".card")`

8. A developer attaches ONE click listener to a parent container instead of a separate listener on every delete button inside it. This technique works because:

   A. click events bubble up from child elements to the container, so it also works for elements added after the page loaded
   B. listeners attached to containers execute faster than listeners attached to buttons
   C. the browser allows only one event listener per page
   D. buttons are not allowed to receive click events directly

9. Which of the following is VALID JSON?

   A. { 'name': 'Ali' }
   B. { "name": "Ali", }
   C. { "name": "Ali" }
   D. { name: "Ali" }

10. `JSON.stringify()` is used to:

    A. convert a JSON string into a JavaScript object
    B. convert a JavaScript object into a JSON string
    C. check whether a JSON document is valid
    D. download JSON data from a server

11. A REST API successfully creates a new resource in response to a POST request. The MOST appropriate HTTP status code to return is:

    A. 200
    B. 201
    C. 400
    D. 404

12. In the chain `fetch(url).then(res => res.json()).then(data => ...)`, a second `.then()` is required because:

A. `res.json()` reads the body asynchronously and returns another Promise
B. `fetch()` always requires exactly two callback functions
C. the first `.then()` only handles network errors
D. JSON data must always be parsed twice for security reasons

13. The three states of a JavaScript Promise are:

A. start, running, finished
B. open, closed, failed
C. pending, fulfilled, rejected
D. synchronous, asynchronous, awaited

14. Consider the program in Figure 1. In which order are the letters printed to the console?

```
console.log("A");
fetch("https://api.example.com/data")
  .then(response => response.json())
  .then(data => console.log("C"));
console.log("B");
```

Figure 1

A. A, C, B
B. A, B, C
C. C, A, B
D. B, C, A

15. Vue.js is best described as:

A. a progressive JavaScript framework for building user interfaces
B. a relational database management system
C. a server-side programming language
D. a web server for hosting PHP applications

16. Which statement about `v-if` and `v-show` is TRUE?

A. `v-show` adds and removes the element from the DOM
B. `v-if` and `v-show` behave identically
C. `v-if` is only used for rendering lists
D. `v-if` adds/removes the element from the DOM, while `v-show` only toggles the CSS `display` property

17. Given `const count = ref(0)` inside `<script setup>`, which statement correctly increases the value in JavaScript code?

A. `count++`
B. `this.count++`
C. `count.value++`
D. `count(1)`

18. In Vue component communication, “props down, events up” means:

A. a parent passes data to a child through props, and the child notifies the parent using emitted events
B. a child component modifies the parent’s props directly
C. events are used to pass data from parent to child
D. all component data must be stored in Pinia

19. Which HTTP request is IDEMPOTENT (calling it several times gives the same result as calling it once)?

A. POST /students
B. PUT /students/3
C. POST /auth/login
D. PATCH /students/3

20. The three parts of a JSON Web Token (JWT), in order, are:

A. username, password, secret
B. issuer, subject, audience
C. signature, payload, header
D. header, payload, signature

SECTION B: STRUCTURED QUESTIONS [45 MARKS]

(INSTRUCTION: This section consists of FOUR (4) questions. Answer ALL questions in the spaces provided.)

QUESTION 1 **[8 MARKS]**

Web Application Architecture

a) With the aid of the *request–response cycle*, describe how a web browser and a web server exchange information for ONE complete page load of a typical web application. Your answer must state the protocol used and the role of the database in the cycle.

(4 marks)

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b) State the role of EACH of the following components of a web application in ONE sentence each:

i) Frontend

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ii) Backend

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iii) Database

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iv) API

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(4 marks)

QUESTION 2 **[10 MARKS]**

Asynchronous JavaScript

a) Define *asynchronous programming* in JavaScript, and explain why the browser does NOT freeze while a `fetch()` request is waiting for the server's response.

(3 marks)

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b) Figure 2 shows a program written using a .then() chain.

```
fetch("https://api.example.com/students/5")
  .then(function (response) {
    return response.json();
  })
  .then(function (student) {
    console.log(student.name);
  })
  .catch(function (error) {
    console.log("Error:", error.message);
  });
```

Figure 2

Rewrite the program in Figure 2 as an async/await function named getStudent() with a try/catch block. Your answer must also throw an Error containing the HTTP status code when response.ok is false.

(5 marks)

c) State ONE advantage and ONE disadvantage of async/await compared with .then() chaining.

(2 marks)

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QUESTION 3 **[12 MARKS]**

Vue.js Frontend Framework

a) Explain what a *frontend framework* is, and state TWO problems of building large applications with plain JavaScript that frameworks solve.

(3 marks)

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b) A Vue 3 project is organised into *Single File Components* (SFC). Name the THREE blocks of an SFC and state the purpose of each block. Your answer must also explain what the `scoped` attribute does.

(4 marks)

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c) Explain why a `ref()` value must be accessed as `count.value` inside `<script setup>`, but is written as `{{ count }}` inside the template.

(2 marks)

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d) Explain the purpose of the `:key` attribute in `v-for`, and why using the array *index* as the key is discouraged when the list can be reordered or items can be deleted.

(3 marks)

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QUESTION 4 **[15 MARKS]**

Backend, Security and Mobile Deployment

a) Explain the concept of a RESTful API. Your answer must describe how resources are named in the URI, and give the mapping of the FOUR main HTTP verbs to their CRUD operations.

(4 marks)

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b) A Vue frontend logs in to a Slim backend that uses *JWT authentication*. Describe the complete authentication flow in FIVE steps, from the user submitting the login form until a protected route returns data. Your answer must state the HTTP status code returned when the token is missing or invalid.

(5 marks)

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c) Name and describe TWO security threats covered in this course, and state ONE prevention technique for EACH threat.

(4 marks)

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d) Explain at a high level how a completed Vue web project becomes an installable Android application using *Capacitor*. Your answer must mention the role of the *WebView*.

(2 marks)

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SECTION C: CODE APPLICATION QUESTIONS [35 MARKS]

(INSTRUCTION: This section consists of THREE (3) questions. Answer ALL questions. Write syntactically correct code in the spaces provided.)

QUESTION 5 **[10 MARKS]**

HTML, CSS and the DOM

a) Write the complete `<body>` content of a "Student Feedback" page using *semantic HTML elements*: a header containing the page heading; a navigation area with TWO links (Home, Feedback); a main content area containing ONE section with a heading and a paragraph; and a footer containing the faculty's contact address.

(4 marks)

b) The class `.card-grid` uses CSS Grid and displays ONE column by default (mobile-first). Write the CSS, including the media query, so that the layout changes to THREE equal columns on screens 1024 px and wider.

(2 marks)

c) Figure 3 shows part of an HTML page.

```
<input id="studentName" type="text">
<button id="addBtn">Add</button>
<ul id="nameList"></ul>
```

Figure 3

Write JavaScript so that when the button is clicked: the input value is read and trimmed; if it is empty, `alert("Name cannot be empty!")` is shown; otherwise a new `<li>` element is created, its text is set using `textContent`, it is appended to the list, and the input is cleared. Do **NOT** use `innerHTML`.

(4 marks)

QUESTION 6 **[10 MARKS]**

Vue Component

Write a COMPLETE Vue 3 Single File Component named `CourseList.vue` using `<script setup>` that satisfies ALL of the following requirements:

i) receives a prop named `title` of type String, required, and displays it inside an `<h2>`;

ii) declares a reactive string `newCourse` and binds it to a text input using `v-model`;

iii) declares a reactive array `courses`; clicking an "Add" button appends the trimmed value of `newCourse` to `courses` and clears the input — an empty value must NOT be added;

iv) renders `courses` as list items using `v-for` with an appropriate `:key`.

(10 marks)

QUESTION 7 **[15 MARKS]**

Full-Stack Backend: Slim 4, PDO and JWT

The faculty is extending the *Books API* from your lab work, built with Slim 4, PDO and MySQL.

a) Write the complete Slim 4 route for `POST /api/books`. The route must: read the JSON request body using `getParsedBody()`; if `title` is missing or empty, return status **400** with the JSON body `{"error":"title is required"}`; otherwise return status **201** with the JSON body `{"message":"Book created"}`. Every response must carry the header `Content-Type: application/json`. Assume the JSON body-parsing middleware is registered and `Request`/`Response` are already imported.

(5 marks)

b) Write PHP code that inserts a new row into the table `books(title, author, year)` using a PDO **PREPARED STATEMENT** with NAMED placeholders. Assume the connection `$pdo` already exists and the values are stored in `$title`, `$author` and `$year`. After the insert, store the new auto-increment id in `$newId`.

(5 marks)

c) Figure 4 shows a vulnerable code fragment from a teammate.

```
$id = $_GET['id'];
$sql = "SELECT * FROM books WHERE id = $id";
$rows = $pdo->query($sql)->fetchAll();
```

Figure 4

i) Name the attack that this code enables, and give ONE example of a malicious input value.

(1 mark)

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ii) Rewrite the code fragment so that it is protected against the attack.

(2 marks)

d) A protected route requires a JWT. State the EXACT HTTP header (name and format) that the client must send with each request, and state the TWO things the middleware verifies before allowing access.

(2 marks)

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END OF QUESTION PAPER